

Cost of Living & Savings Data Visualization Project

1. Project Purpose

This project uses a synthetic cost-of-living dataset to explore how far a typical paycheck can stretch across different countries and years.

The main audience is students and young professionals who are considering which country to live or work in.

Instead of only comparing which country has the highest salary, the project focuses on what happens to income after tax and essential expenses.

The core questions are:

- After paying tax, housing, healthcare, education, and transportation, how much room is left for saving?
- How does this balance between spending and saving differ across countries and regions?
- Has it become easier or harder to save over time in different countries?

2. Data and Derived Metrics

The raw dataset contains, for each Country and Year:

- Average_Monthly_Income
- Cost_of_Living (index)
- Tax_Rate
- Housing_Cost_Percentage
- Savings_Percentage
- Healthcare_Cost_Percentage
- Education_Cost_Percentage
- Transportation_Cost_Percentage
- Region

From these variables, several derived metrics are created in Python to support analysis and visualization:

1) TakeHome (after-tax income)

$\text{TakeHome} = \text{Average_Monthly_Income} * (1 - \text{Tax_Rate} / 100).$

This represents the income that remains after tax and is the basis for evaluating how income is allocated.

2) Housing_Share_TakeHome_Percent

Percentage of net (after-tax) income spent on housing:

$\text{Housing_Share_TakeHome_Percent} = (\text{Housing_Cost_Percentage} / (100 - \text{Tax_Rate})) * 100.$

3) Savings_Share_TakeHome_Percent

Percentage of net income allocated to savings:

$\text{Savings_Share_TakeHome_Percent} = (\text{Savings_Percentage} / (100 - \text{Tax_Rate})) * 100.$

4) Net_Savings_Room_Percent

A simple "net savings space" metric:

$\text{Net_Savings_Room_Percent} = \text{Savings_Share_TakeHome_Percent} - \text{Housing_Share_TakeHome_Percent}.$

Positive values mean a larger share of net income goes into savings than into housing; negative values indicate heavy housing pressure.

5) Other_Essentials_Percentage

The remaining share of income spent on essential categories other than tax, housing, and savings:

$\text{Other_Essentials_Percentage} = 100 - (\text{Tax_Rate} + \text{Housing_Cost_Percentage} + \text{Savings_Percentage}$

+

$\text{Healthcare_Cost_Percentage} + \text{Education_Cost_Percentage} + \text{Transportation_Cost_Percentage}).$

6) Savings_to_Spending_Ratio_Net

A ratio describing how much is saved for every unit of net income that is spent:

Let $s_{\text{net}} = \text{Savings_Share_TakeHome_Percent} / 100$.

Then $\text{Savings_to_Spending_Ratio_Net} = s_{\text{net}} / (1 - s_{\text{net}})$.

For example, a value of 0.4 means that for every 1 unit of net income spent, 0.4 units are saved.

These metrics are exported into a processed CSV file (`cost_of_living_with_metrics_percent.csv`) and used directly in Tableau.

3. Pivot Transformation for Cost Components

To visualize the internal composition of cost of living using a donut chart in Tableau, the project pivots five cost components:

- Tax_Rate
- Housing_Cost_Percentage
- Healthcare_Cost_Percentage
- Education_Cost_Percentage
- Transportation_Cost_Percentage

In Tableau's Data Source pane, these five columns are selected and pivoted. The pivot operation produces two new fields:

- Cost_Component: contains the name of the component (Tax_Rate, Housing, Healthcare, Education, Transportation).
- Cost_Component_Percent: contains the corresponding percentage value for that component.

After the pivot, each Country–Year combination has five rows, one for each component. This structure is ideal for building a donut chart:

- Cost_Component is placed on Color,
- Cost_Component_Percent is placed on Angle and Label,

and a dual-axis technique with an inner white pie is used to create the donut shape.

The resulting donut chart answers the question:

"Within the set of essential cost components (tax, housing, healthcare, education, transportation), where does this country spend the largest share of income in a given year?"

4. Intended Visualizations and Insights

The processed dataset and pivot transformation support several key visualizations in Tableau:

- A world map with a year slider (2000–2023) that colors each country by `Net_Savings_Room_Percent`, showing where it is easier or harder to save in a given year.
- A donut chart that shows the internal structure of essential cost components for the selected country and year.
- Line charts tracking `Net_Savings_Room_Percent` or `Savings_to_Spending_Ratio_Net` over time for a chosen country, revealing whether saving space is improving or deteriorating.
- Scatter plots comparing TakeHome (after-tax income) with `Net_Savings_Room_Percent` or `Savings_to_Spending_Ratio_Net`, highlighting countries that are high-income but still under strong cost pressure, or moderate-income but relatively favorable for saving.

Overall, the goal of the project is to move beyond simple salary comparisons and provide a more realistic picture of affordability by analyzing how after-tax income is distributed across essential living costs and savings for different countries and years.